MULTIVARIATE DEPTH-BASED DETECTION OF HIDDEN METABOLIC RISK

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ABSTRACT

Statistical depth functions provide a center outward ordering of multivariate data and offer a principled way to measure the extremeness of observations. In this project, we study depth based methods for outlier detection, focusing on the classical Mahalanobis depth and the Kernelized Spatial Depth (KSD). While Mahalanobis depth is widely used due to its simplicity, it relies on strong distributional assumptions and is limited in capturing complex, non elliptical data structures.

To address these limitations, we adopt the KSD framework, which extends spatial depth through positive definite kernels, enabling the detection of nonlinear patterns in the data. We apply both methods to real world data from multiple cycles of the National Health and Nutrition Examination Survey (NHANES), restricting attention to adults with normal body mass index(BMI) to investigate hidden metabolic risk.

Our analysis shows that KSD consistently identifies a subset of individuals with significantly higher HbA1c levels compared to those identified exclusively by Mahalanobis depth. This difference is statistically validated using one sided Wilcoxon rank sum tests across all cycles ($p < 0.05$), indicating that KSD captures clinically meaningful outliers that are not detected by traditional distance based methods. These findings highlight the advantage of kernel based depth measures in uncovering subtle but important deviations in real world data and suggest their potential utility in improving risk screening frameworks.

Keywords Kernelized Spatial Depth · Mahalanobis Distance · Outlier Detection

1 Introduction

Outlier detection is a fundamental problem in statistical analysis, with applications ranging from anomaly detection to robust inference. A common approach to this problem is through statistical depth functions, which provide a center outward ordering of multivariate data. In this project, we study depth based methods for outlier detection, focusing in particular on Mahalanobis depth and the Kernelized Spatial Depth (KSD) function.

We begin with a review of the literature, primarily based on the work of Yixin Chen, Xin Dang, Hanxiang Peng, and Henry L. Bart Jr., which introduces the KSD framework as a flexible and robust alternative to classical depth measures [Chen09]. While Mahalanobis depth is widely

used due to its simplicity and analytical tractability, it relies heavily on distributional assumptions and can perform poorly in non elliptical or complex data settings.

To address these limitations, we explore the Kernelized Spatial Depth, which incorporates kernel methods to better capture nonlinear structures in the data. Finally, we apply both depth measures to a real dataset, demonstrating the advantages of KSD over Mahalanobis depth in identifying outliers in more general settings.

2 Outlier

The problem of outlier detection can be stated as follows: given observations $X_1, \ldots, X_n \in \mathbb{R}^d$ drawn from an unknown distribution P, possibly contaminated by a small fraction of observations from another distribution Q, identify the subset of indices corresponding to observations that do not conform to P. Formally, under a contamination model

$$X_i \sim (1-\varepsilon)P + \varepsilon Q, \quad 0 < \varepsilon \ll 1,$$

the task is to decide, based only on the sample, which X_i's are generated from Q. The difficulty lies in the absence of a canonical definition of extremeness, especially in higher dimensions where geometric intuition and distance based separation become less reliable.

The intuitive need for a depth function arises precisely from this lack of ordering in $\mathbb{R}^d$. In one dimension, outliers can be identified using the natural ordering of the real line, but in higher dimensions no such total order exists. A depth function provides a center outward ranking of points, assigning higher values to central observations and lower values to peripheral ones. This allows us to define outliers as points with sufficiently small depth, thereby giving a principled, geometry aware notion of extremeness that generalizes univariate ideas like medians, quantiles, and tail behavior. [MV25]

3 What is Data Depth?

Data depth is a functional $D(x; P)$ that assigns to each point $x \in \mathbb{R}^d$ a nonnegative real number measuring how "central" x is with respect to a probability distribution P . Formally, a depth function is a map

$$D : \mathbb{R}^d \times \mathcal{P}(\mathbb{R}^d) \to [0, 1],$$

where larger values indicate greater centrality and smaller values indicate outlyingness.

A well defined depth function typically satisfies the following axioms :

1. **Affine invariance:** For any nonsingular affine transformation $T(x) = Ax + b$,

$$D(x; P) = D(Tx; P_T),$$

   where P_T is the pushforward of P.

2. **Maximality at center:** If P has a center of symmetry θ, then

$$D(\theta; P) = \sup_x D(x; P).$$

3. **Monotonicity relative to deepest point:** Along any ray from a deepest point θ, depth decreases monotonically.

4. **Vanishing at infinity:**

$$D(x; P) \to 0 \quad \text{as } \|x\| \to \infty.$$

Now there are several types of data depth. Some specific types of data depths will be discussed in this report.

Distance based depth functions quantify the centrality of a point $x \in \mathbb{R}^d$ by measuring its distance from a representative location of the distribution, typically a center such as the mean, median, or another robust estimator. A general form is

$$D(x; P) = g\big(d(x, \theta(P))\big),$$

where $d(\cdot, \cdot)$ is a metric $\theta(P)$ is a location functional, and g is a decreasing function ensuring that depth decreases as distance increases.

Mahalanobis depth, L_2depth and Oja depth are a few examples of distance based Depth function.

The Mahalanobis depth of a point $x \in \mathbb{R}^d$ with respect to a distribution P having mean vector μ and positive definite covariance matrix Σ is defined as

$$D(x; P) = \Big(1 + (x - \mu)^\top \Sigma^{-1} (x - \mu)\Big)^{-1}.$$

Points closer to the center μ have higher depth, while points farther away have lower depth. This depth is affine invariant when μ and Σ are taken as the mean and covariance of P, but it can be sensitive to outliers unless robust estimates of location and scatter are used. [BFK10]

Kernel based depth functions can be defined at the population level using a probability distribution P by replacing the empirical average with an expectation. Specifically, for a kernel $K : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$, the depth of a point $x \in \mathbb{R}^d$ is given by

$$D(x; P) = \mathbb{E}_{X \sim P}[K(x, X)] = \int_{\mathbb{R}^d} K(x, y)\, dP(y).$$

Here, K is said to be a positive definite kernel if for any $n \in \mathbb{N}$, any points $x_1, \ldots, x_n \in \mathbb{R}^d$, and any real coefficients $c_1, \ldots, c_n$, one has

$$\sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j K(x_i, x_j) \geq 0.$$

This formulation measures the centrality of x through its average similarity to the distribution P, assigning higher depth to points lying in regions of high probability mass.

Kernelized spatial depth extends the classical spatial depth by incorporating a positive definite kernel to handle nonlinear structure in the data. Let $K(\cdot, \cdot)$ be a positive definite kernel with associated feature map $\phi : \mathbb{R}^d \to \mathcal{F}$, where $\mathcal{F}$ is a feature space such that $K(x, y) = \langle \phi(x), \phi(y) \rangle$. The kernelized spatial depth of a point $x \in \mathbb{R}^d$ with respect to a distribution P is defined as

$$D(x; P) = 1 - \left\| \mathbb{E}_{X \sim P} \left[\frac{\phi(x) - \phi(X)}{\|\phi(x) - \phi(X)\|} \right] \right\|.$$

The sample version replaces the expectation with an average over the sample $X_1, \ldots, X_n$.

$$D(x; P_n) = 1 - \left\| \frac{1}{n} \sum_{i=1}^{n} \frac{\phi(x) - \phi(X_i)}{\|\phi(x) - \phi(X_i)\|} \right\|.$$

Using $\|\phi(x) - \phi(X_i)\|^2 = K(x,x) + K(X_i, X_i) - 2K(x, X_i)$, it can be easily shown that

$$D(x; P_n) = 1 - \left[\frac{1}{n^2} \sum_{i=1}^{n} \sum_{j=1}^{n} \frac{K(x,x) - K(x,X_i) - K(x,X_j) + K(X_i,X_j)}{\sqrt{K(x,x) + K(X_i,X_i) - 2K(x,X_i)}\,\sqrt{K(x,x) + K(X_j,X_j) - 2K(x,X_j)}} \right]^{1/2}.$$

A common choice for $K(x,y)$ is the Gaussian kernel. The Gaussian kernel is defined as

$$K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2} \right),$$

where $x, y \in \mathbb{R}^d$ and $\sigma > 0$ is the bandwidth parameter.

4 Algorithmic aspects

In outlier detection, the false alarm probability (also called the false positive rate) is the probability of incorrectly classifying a normal observation (i.e., one generated from the nominal distribution P or F_{good}) as an outlier. Formally, if $\mathcal{D}$ denotes the decision rule, the false alarm probability is

$$\mathbb{P}_{X \sim P}\big(\mathcal{D}(X) = \text{outlier}\big).$$

Controlling this probability is crucial in practice, as overly aggressive detection rules may flag many genuine observations as anomalies. Typically, the threshold parameter t in depth-based methods is chosen to balance false alarms and missed detections, often by fixing an acceptable false alarm level and calibrating t accordingly.

The input is a collection of observations $\mathcal{X} = \{x_1, x_2, \ldots, x_\ell\} \subset \mathbb{R}^n$, a kernel κ, and a parameter t. The following pseudocodes determine whether an observation x is an outlier. In terms of the number of kernel evaluations and multiplications, the cost of computing the KSD for a given observation is $O(\ell^2)$.

Algorithm 1 Learning an outlier detector [Chen09]

for every pair of x_i and x_j in $\mathcal{X}$ **do**
  $K_{ij} = \kappa(x_i, x_j)$
end for
Given input x
for every observation x_i in $\mathcal{X}$ **do**
  $\zeta_i = \kappa(x, x_i)$
  $\delta_i = \sqrt{\kappa(x,x) + K_{ii} - 2\zeta_i}$
  if $\delta_i = 0$ **then**
    $z_i = 0$
  else
    $z_i = \frac{1}{\delta_i}$
  end if
end for
for every pair of x_i and x_j in $\mathcal{X}$ **do**
  $\widetilde{K}_{ij} = \kappa(x,x) + K_{ij} - \zeta_i - \zeta_j$
end for
$D_\kappa(x, \mathcal{X}) = 1 - \frac{1}{|\mathcal{X} \cup \{x\}| - 1} \sqrt{\mathbf{z}^T \widetilde{K} \mathbf{z}}$
Output: x is an outlier if $D_\kappa(x, \mathcal{X}) \leq t$

5 Statistical test

Wilcoxon Rank-Sum test(one sided) - The rank-sum test, introduced by Frank Wilcoxon and developed by Henry B. Mann and Donald R. Whitney, is a nonparametric method for comparing two independent samples.

Let $X_1, X_2, \ldots, X_m \sim F_X$ and $Y_1, Y_2, \ldots, Y_n \sim F_Y$. Combine all observations and take the sum of the ranks of X sample only, call it T_X.

$H_0 : F_X = F_Y \quad H_1 : P(X > Y) > 1/2 \quad (F_X$ stochastically greater than $F_Y)$

Under H_0: $\mathbb{E}[T_X] = m(m+n+1)/2$ and Var$(T_X) = mn(m+n+1)/12$

In case of small samples, we use the exact distribution of T_X, and reject H_0 for large values of T_X. [Mood74]

6 Dataset

Data is obtained from the National Health and Nutrition Examination Survey (NHANES), conducted by the National Center for Health Statistics under the Centers for Disease Control and Prevention. NHANES is a U.S. representative survey based on a stratified multistage sampling design. We use the following datasets – NHANES(2021-2023), NHANES(2017-2018) and NHANES(2015-2016). From the datasets, a subset of features have been selected for our study. The following features have been selected(refer to Table 1).

Filtering: We restrict to individuals aged ≥ 18 with normal BMI ($18.5 \leq$ BMI ≤ 25), and perform complete case analysis after restricting to the fasting subsample where required. The samples with missing values in any of the features have been deleted.

Procedures on Dataset

- Each NHANES wave (2015–2016, 2017–2018, 2021–2023) is analyzed separately to preserve temporal structure.
- All continuous variables—fasting glucose, HbA1c, triglycerides, and HDL—are standardized using z-score normalization to prevent scale dominance in distance based computations.
- We then perform outlier detection using Mahalanobis Distance of datapoints from the mean vector of the data. We flagged outliers as data points exceeding the 95th percentile of the Mahalanobis Distance (MD). This is equivalent to taking the bottom 5 percentile of the Mahalanobis depth—calculated as $1/(1 + \mathrm{MD}^2)$—meaning both approaches yield identical outliers.
- Next, we perform Kernelized spatial depth computation of datapoints. For the purpose we have chosen the Gaussian kernel as it induces a similarity based embedding, where the bandwidth σ controls the local–global trade off: smaller σ enhances local sensitivity, while larger σ yields a more global representation through aggregated pairwise similarities. We flag top 5 percentile of least KSD as outliers.
- Finally, we compare the sets of outliers detected by KSD outlier detection algorithm and MD outlier detection algorithm.
- To assess clinical relevance of disagreement between the two detectors, we focus on HbA1c levels within observations uniquely identified by Kernelized Spatial Depth method (KSD only) and those uniquely identified by Mahalanobis distance method (MD only). We then apply a one-sided Wilcoxon rank sum test to evaluate whether HbA1c values in the KSD only group are stochastically larger than those in the

MD only group. This tests the hypothesis that KSD selectively identifies individuals with comparatively higher glycemic burden among discordant outliers.

- The outliers for all three cohorts are plotted after PCA(PC1 vs PC2 plot).
- The R codes are attached in the appendix, Also check for the required files in [Proj25]

7 Results and Interpretation

Outliers were identified using both Mahalanobis Distance (MD) and Kernelized Spatial Depth (KSD). For each dataset, we compare observations uniquely detected by each method. A one sided Wilcoxon Rank Sum Test was used to assess differences in HbA1c:

$$H_0 : \text{HbA1c}_{\text{KSD}} = \text{HbA1c}_{\text{MD}}, \quad H_1 : \text{HbA1c}_{\text{KSD}} >_{st} \text{HbA1c}_{\text{MD}} \tag{1}$$

2015–2016 (n = 626)

Total outliers = 32; Outliers unique to each method = 4.
Means (Fasting Glucose(mg/dL), HbA1c(%), Triglyceride(mg/dL), HDL cholesterol(mg/dL):

- **KSD:** (169, 8.20, 113, 59.8)
- **MD:** (104, 6.62, 90.5, 74.8)

$p = 0.014 \Rightarrow$ Reject H_0

2017–2018 (n = 639)

Total outliers = 32; Outliers unique to each method = 6.
Means (Fasting Glucose(mg/dL), HbA1c(%), Triglyceride(mg/dL), HDL cholesterol(mg/dL):

- **KSD:** (175, 7.38, 136, 52.5)
- **MD:** (106, 5.05, 127, 85)

$p = 0.002 \Rightarrow$ Reject H_0

2021–2023 (n = 849)

Total outliers = 43; Outliers unique to each method = 6.
Means (Fasting Glucose(mg/dL), HbA1c(%), Triglyceride(mg/dL), HDL cholesterol(mg/dL):

- **KSD:** (159, 7.35, 152, 42.5)
- **MD:** (101, 5.03, 104, 96.3)

$p = 0.002 \Rightarrow$ Reject H_0

Overall Interpretation

Across all datasets, HbA1c values among KSD identified outliers are consistently higher than those identified by MD, with statistically significant differences. This indicates that KSD preferentially captures more clinically extreme (high HbA1c) observations compared to MD. Please refer to Table 2 for details.

8 Conclusion

The results show that outlier detection using kernelized spatial depth performs better than Mahalanobis distance (MD) in capturing the true structure of the data, particularly in non elliptical or complex distributions. While MD relies on global covariance structure and assumes approximate elliptical symmetry, it fails to correctly identify outliers when the data exhibits non linear shapes or multiple clusters. In contrast, kernelized spatial depth adapts to local geometry, allowing it to detect outliers that are structurally isolated even if they are not far in terms of global distance.

Empirical observations confirm that MD can misclassify central points in non standard distributions as normal due to its rigid contour structure, whereas kernelized spatial depth provides more flexible contours aligned with the data. This leads to more reliable identification of outliers in practical scenarios.

Overall, kernelized spatial depth offers a clear advantage over MD for datasets with complex or non elliptical structure, while MD remains limited to simpler settings where its underlying assumptions hold.

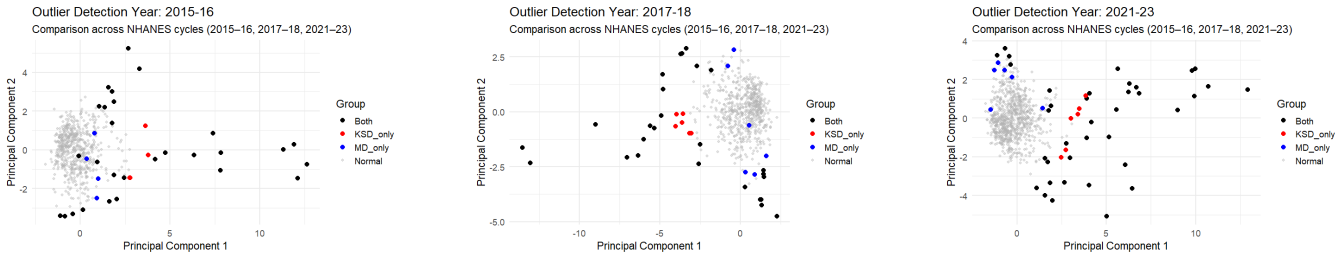
9 Plots and Tables

Feature code	**Feature**	**Description**	**Normal reference**
SEQN	Identifier	Sequence number of an individual	–
RIDAGEYR	Age	Age of individuals	–
BMXBMI	BMI ($\mathrm{kg/m^2}$)	Weight (kg) divided by height (m)^2	18.5–24.9: normal
LBXGLU	Fasting glucose (mg/dL)	Blood glucose level after at least 8 hours of fasting	< 100: normal
LBXGH	HbA1c (%)	Glycated hemoglobin; reflects average blood glucose over 2–3 months	$< 5.7\%$: normal
LBXTLG	Triglycerides (mg/dL)	Fasting triglyceride concentration in blood sample	< 150: normal
LBXHDD	HDL cholesterol (mg/dL)	High density lipoprotein ("good cholesterol")	≥ 60: normal

Table 1

Year	Data Clean	MD/KSD Outliers	MD/KSD only Outliers		Glucose	Hba1c	Triglycerides	HDL
2015-16	626	32	4	KSD	169	8.2	113	59.8
				MD	104	6.62	90.5	74.8
2017-18	639	32	6	KSD	175	7.38	136	52.5
				MD	106	5.05	127	85
2021-23	849	43	6	KSD	159	7.35	152	42.5
				MD	101	5.03	104	96.3

Table 2: Mean Expression of different Markers for KSD and MD only outliers



(a) (b) (c)

Figure 1: Comparison across NHANES cycles (2015-16, 17-18, 21-23)

Appendix: R code

```
# DATA_EXTRACTION_AND_OUTLIER_DETECTION

library(haven)
library(dplyr)
library(tidyr)
library(dbscan)
library(beepr)

prefix <- "17_18_" # change to other prefixes as well
demo <- read_xpt(paste0(prefix, "DEMO_L.xpt"))
bmx  <- read_xpt(paste0(prefix, "BMX_L.xpt"))
glu  <- read_xpt(paste0(prefix, "GLU_L.xpt"))
ghb  <- read_xpt(paste0(prefix, "GHB_L.xpt"))
trig <- read_xpt(paste0(prefix, "TRIGLY_L.xpt"))
hdl  <- read_xpt(paste0(prefix, "HDL_L.xpt"))
dim(demo)
dim(bmx)
dim(glu)
dim(ghb)
dim(trig)
dim(hdl)
data <- demo %>%
  select(SEQN, RIDAGEYR) %>%
  left_join(bmx %>% select(SEQN, BMXBMI), by = "SEQN") %>%
  left_join(glu %>% select(SEQN, LBXGLU), by = "SEQN") %>%
  left_join(ghb %>% select(SEQN, LBXGH), by = "SEQN") %>%
  left_join(
    trig %>%
      select(SEQN, LBXTLG) %>%
      rename(LBXTLG = LBXTLG),
    by = "SEQN"
  ) %>%
  left_join(hdl %>% select(SEQN, LBDHDD), by = "SEQN")
data_clean <- data %>%
  filter(RIDAGEYR >= 18,
         BMXBMI >= 18.5, BMXBMI < 25) %>%
  drop_na()
dim(data_clean)
X <- data_clean %>%
  select(LBXGLU, LBXGH, LBXTLG, LBDHDD, RIDAGEYR)
X <- scale(X)
X_mat <- as.matrix(X)
dim(X_mat)


#md
md <- mahalanobis(X_mat, colMeans(X_mat), cov(X_mat))
threshold_md <- quantile(md, 0.95)
```

```
out_md <- md > threshold_md
sum(out_md)

#ksd
dists <- as.matrix(dist(X_mat))
sigma <- median(dists)
K <- exp(-(dists^2) / (2 * sigma^2))

n <- nrow(X_mat)
ksd_exact <- numeric(n)
cat("\nStarting EXACT KSD...\n")
start_time <- Sys.time()
for (i in 1:n) {
  elapsed <- as.numeric(difftime(Sys.time(), start_time, units = "mins"))
  cat(sprintf("i=%d/%d (%.1f%%) | %.2f min\r", i, n, 100*i/n, elapsed))
  s_val <- 0
  for (j in 1:n) {
    for (k in 1:n) {
      num <- K[i,i] + K[j,k] - K[i,j] - K[i,k]
      denom1 <- sqrt(K[i,i] + K[j,j] - 2*K[i,j])
      denom2 <- sqrt(K[i,i] + K[k,k] - 2*K[i,k])
      if (denom1 > 0 && denom2 > 0) {
        s_val <- s_val + num / (denom1 * denom2)
      }
    }
  }

  ksd_exact[i] <- 1 - sqrt(s_val / (n^2))
}
cat("\nKSD completed in:",
    difftime(Sys.time(), start_time, units = "mins"), "minutes\n")
beepr::beep()

# KSD OUTLIERS
threshold_ksd <- quantile(ksd_exact, 0.05)
out_ksd <- ksd_exact < threshold_ksd

# INDEX SETS
idx_md  <- which(out_md)
idx_ksd <- which(out_ksd)
ksd_only <- setdiff(idx_ksd, idx_md)
md_only  <- setdiff(idx_md, idx_ksd)
common   <- intersect(idx_md, idx_ksd)


#verification:
cat("MD:", length(idx_md), "\n")
cat("KSD:", length(idx_ksd), "\n")
cat("Overlap:", length(common), "\n")
cat("KSD only:", length(ksd_only), "\n")
cat("MD only:", length(md_only), "\n")


#analysis
analyze <- function(idx, name, df) {
  cat(paste("\n---", name, "---\n"))
  print(
    df[idx, ] %>%
      summarise(
        glucose = mean(LBXGLU),
        hba1c   = mean(LBXGH),
        trig    = mean(LBXTLG),
        hdl     = mean(LBDHDD)
      )
  )
}
analyze(ksd_only, "KSD only", data_clean)
```

```
analyze(md_only, "MD only", data_clean)

# TESTING_AND_PLOTTING

library(dplyr)
library(ggplot2)
results <- list()
process_cycle <- function(prefix) {
  # LOAD
  data_clean <- readRDS(paste0(prefix, "data_clean.rds"))
  ksd_exact  <- readRDS(paste0(prefix, "ksd_exact.rds"))
  out_md     <- readRDS(paste0(prefix, "out_md.rds"))
  # KSD
  threshold_ksd <- quantile(ksd_exact, 0.05)
  out_ksd <- ksd_exact < threshold_ksd
  # INDICES
  idx_md  <- which(out_md)
  idx_ksd <- which(out_ksd)
  ksd_only <- setdiff(idx_ksd, idx_md)
  md_only  <- setdiff(idx_md, idx_ksd)
  common   <- intersect(idx_md, idx_ksd)
  ksd_values <- data_clean$LBXGH[ksd_only]
  md_values  <- data_clean$LBXGH[md_only]
  wilcox_test <- wilcox.test(
    ksd_values,
    md_values,
    alternative = "greater",
    paired = FALSE
  )
  summarize_group <- function(idx) {
    data_clean[idx, ] %>%
      summarise(
        glucose = mean(LBXGLU),
        hba1c   = mean(LBXGH),
        trig    = mean(LBXTLG),
        hdl     = mean(LBDHDD)
      )
  }
  list(
    n = nrow(data_clean),
    ksd_only_n = length(ksd_only),
    md_only_n  = length(md_only),
    common_n   = length(common),
    ksd_stats = summarize_group(ksd_only),
    md_stats  = summarize_group(md_only),
    wilcox_pvalue = wilcox_test$p.value,

    ksd_only_idx = ksd_only,
    md_only_idx  = md_only
  )
}
results[["15_16"]] <- process_cycle("15_16_")
results[["17_18"]] <- process_cycle("17_18_")
results[["21_23"]] <- process_cycle("21_23_")

data.frame(
  Cycle = c("15-16","17-18","21-23"),
  KSD_HbA1c = c(
    results[["15_16"]]$ksd_stats$hba1c,
    results[["17_18"]]$ksd_stats$hba1c,
    results[["21_23"]]$ksd_stats$hba1c
  ),
  MD_HbA1c = c(
    results[["15_16"]]$md_stats$hba1c,
    results[["17_18"]]$md_stats$hba1c,
    results[["21_23"]]$md_stats$hba1c
  ),
```

```
  Wilcox_pvalue = c(
    results[["15_16"]]$wilcox_pvalue,
    results[["17_18"]]$wilcox_pvalue,
    results[["21_23"]]$wilcox_pvalue
  )
)

results[["15_16"]]$wilcox_pvalue
results[["17_18"]]$wilcox_pvalue
results[["21_23"]]$wilcox_pvalue

prefix <- "21_23_"   # change to other prefixes as well
data_clean <- readRDS(paste0(prefix, "data_clean.rds"))
ksd_exact  <- readRDS(paste0(prefix, "ksd_exact.rds"))
out_md     <- readRDS(paste0(prefix, "out_md.rds"))
threshold_ksd <- quantile(ksd_exact, 0.05)
out_ksd <- ksd_exact < threshold_ksd
idx_md  <- which(out_md)
idx_ksd <- which(out_ksd)
ksd_only <- setdiff(idx_ksd, idx_md)
md_only  <- setdiff(idx_md, idx_ksd)
common   <- intersect(idx_md, idx_ksd)
X <- data_clean %>%
  select(LBXGLU, LBXGH, LBXTLG, LBDHDD, RIDAGEYR) %>%
  scale()
pca <- prcomp(X)
df <- as.data.frame(pca$x[,1:2])
colnames(df) <- c("PC1", "PC2")
df$group <- "Normal"
df$group[ksd_only] <- "KSD_only"
df$group[md_only]  <- "MD_only"
df$group[common]   <- "Both"
ggplot(df, aes(PC1, PC2, color = group)) +
  geom_point(
    data = subset(df, group == "Normal"),
    alpha = 0.4, size = 1.2
  ) +
  geom_point(
    data = subset(df, group != "Normal"),
    size = 2.5
  ) +
  scale_color_manual(values = c(
    "Normal" = "grey70",
    "KSD_only" = "red",
    "MD_only" = "blue",
    "Both" = "black"
  )) +
  labs(
    title = paste("Outlier Detection Year: ", sep = ""),
    subtitle = "Comparison across NHANES cycles (2015-16, 2017-18, 2021-23)",
    x = "Principal Component 1",
    y = "Principal Component 2",
    color = "Group"
  ) +
  theme_minimal(base_size = 14)
```

References

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